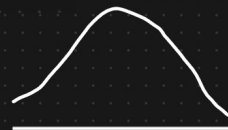


① Log Normal Distribution

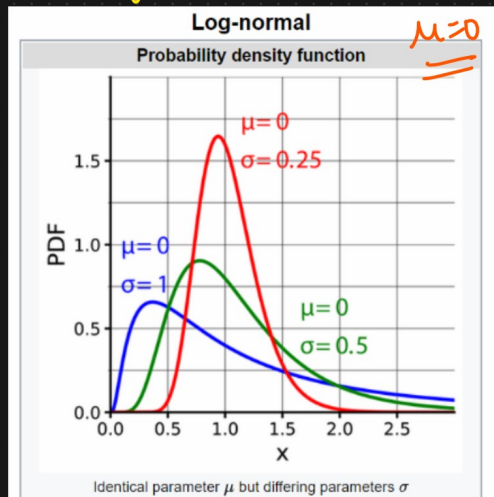


Empirical Rule

In probability theory, a log-normal (or lognormal) distribution is a continuous probability distribution of a random variable whose logarithm is normally distributed. Thus, if the random variable X is log-normally distributed, then $Y = \ln(X)$ has a normal distribution. Equivalently, if Y has a normal distribution, then the exponential function of Y , $X = \exp(Y)$, has a log-normal distribution

Right Skewed Distribution

Assume X is random variable which belongs to log normal distribution.



$$X \sim \text{Log Normal Distribution}(\mu, \sigma)$$



$$Y \sim \ln(X) = \text{Normal Distribution}$$



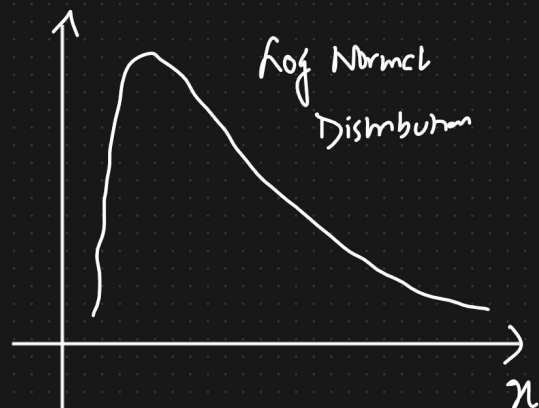
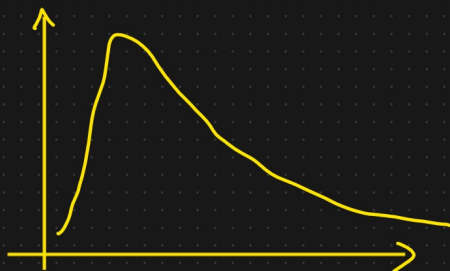
Natural log
[log_e]



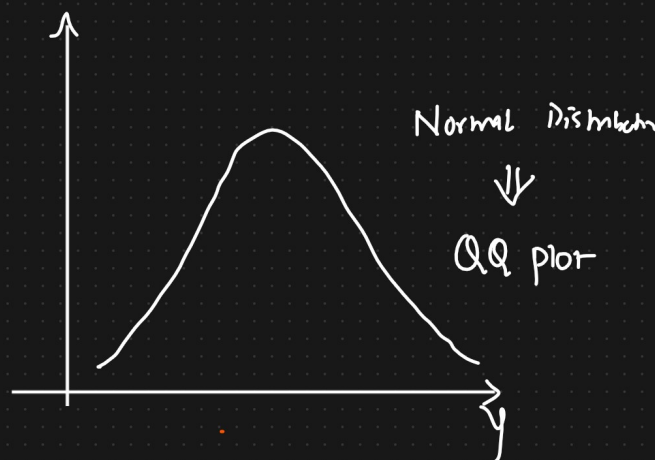
vice versa

$$X \sim \exp(Y) \Rightarrow \text{Log Normally Distributed}$$

See the curve at right side bcs it is something like parallel to x-axis that's why



$$\ln(x) \Rightarrow \exp(y)$$



Normal Distribution

↓
QQ plot

Eg:

- ① Wealth distribution of the world
- ② Discussion Forum $\rightarrow$ Length of the comments
- ③ Length of chess games
- ④ Dwell time on online articles (joke, news)
- ⑤ Salaries of employees in a company.

